

A novel concept for integrating and delivering automobile warranty reliability information via a comprehensive digital dashboard

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Abstract

This study introduces a novel digital dashboard aimed at reforming warranty prediction and quality improvement in the automotive sector. By bridging the gap identified in existing research on statistical modeling, machine learning, and reliability analysis tools, this paper offers a practical solution tailored to the needs of field quality engineers. The dashboard facilitates rapid, data-driven decisions with minimal data requirements, addressing the complexities of warranty claims analysis, cost management, and the enhancement of product quality. Developed through collaboration with industry experts and iterative testing, it combines user-friendly interfaces with advanced algorithms for reliability analysis, warranty cost forecasting, and improvement measure validation. The implementation of this dashboard demonstrates significant improvements in operational efficiency, error reduction, and cost savings, marking a pivotal step towards sustainable quality enhancement and customer satisfaction in the automotive industry.

Keywords: Warranty claim, Reliability, Dashboard, Prediction, Automobile

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1. Introduction

Ensuring high quality is crucial for corporate profitability and customer satisfaction. The foundation for achieving high quality starts with accurate and effective claims analysis. Properly analyzing claims allows for the identification and improvement of product defects, thereby increasing customer satisfaction and preventing unnecessary costs. Dashboards, through data analysis, play a pivotal role in predicting claim trends and facilitating swift and accurate responses to maintain service quality and optimize costs. This paper

specifically focuses on the design of a digital dashboard from the perspective of automotive components, highlighting important factors, features, and the impact on effective claim analysis and management.

Warranty is an agreement based on a contract or arrangement between consumers and sellers or manufacturers regarding the quality and performance of a product or service, including legal regulations. Accordingly, manufacturing or selling companies are obligated to address malfunctions occurring within the warranty period (Blischke et al., 2011). Products of poor quality incur not only high warranty costs but also additional associated expenses. Generally, having a longer warranty period than competing companies can potentially boost sales, but it may also result in additional costs (Dai et al., 2019). In order to minimize additional costs, it is necessary to provide an appropriate warranty period according to the warranty cost for each part (Yang et al., 2023). Furthermore, in order to maximize profit, the selection of not only the warranty period but also the items covered by free maintenance and their duration is necessary (Ruan et al., 2022). Warranty cost management is essential for financial stability, budget planning and management, cost optimization, customer satisfaction, legal compliance, corporate image and credibility, and risk management. Two key measures of warranty costs are warranty liability and warranty claims costs. The warranty liability ratio represents the proportion of reserve funds set aside by a company for product warranties each time a product is sold, used to provide warranty services during the warranty period. Warranty claims ratio is measured by the frequency at which customers actually request and receive warranty services for a product. A high claims ratio may indicate potential quality issues with the product, leading to additional costs for the company.

In the global automotive industry of 2022, major companies exhibit an average warranty liability ratio ranging from 1% to 5%, resulting in costs amounting to approximately \$54.655 billion. Furthermore, the actual warranty claims ratio falls within the range of 1% to 4%, with associated costs reported at \$43.14 billion (Warrantyweek, 2023). Reducing warranty claims costs involves minimizing the likelihood of consumer failures within the warranty period. Consumers tend to evaluate products with higher failure probabilities as having significantly lower value. If a product is perceived as prone to malfunctions or of inferior quality, the consumer's willingness to pay decreases (Cabral, 2007; Grewal et al., 1994; Teo and Yeong, 2003; Jiang et al., 2017; Wang et al., 2017). Objective quality refers to measurements obtained

through evaluations by institutions and experts, while perceived quality is a subjective judgment based on expectations about the quality, incorporating one's own and others' experiences, and considering various sources such as brand reputation, price, and advertising (Mitra and Golder, 2006). When it comes to product failures during the warranty period, they are directly associated with objective quality and ultimately impact customers' perceived quality. In other words, reducing warranty costs not only brings financial benefits but also influences consumers' willingness to repurchase, making it highly relevant to a company's sustainability and viability.

To reduce warranty claims costs, it is essential to start with an analysis of warranty data. Identifying components that incur significant costs and analyzing whether these components will experience a surge or decline in failures in the future are vital steps in formulating production and field strategies. However, accurately predicting warranty claims remains a challenging task for many companies. The reason for this difficulty lies in the imperfect nature of warranty data for statistical analysis. Warranty data is often incomplete due to the limited duration of the warranty period, and it may suffer from data contamination, inaccuracy, a spike in claims at the end of the warranty period, and the continuous evolution of warranty indices such as Warranty growth, which includes factors like claims per 1000 units (R/1000), Cost Per Unit (CPU). Additionally, predicting warranty maturity is inaccurate due to challenges such as the limited early-failure data during the initial shipment and sampling limitations. These challenges highlight the complexity of accurately prediction warranty claims (Rai and Singh, 2009).

Fig. 1 illustrates an example flow of reliability analysis using Minitab after preprocessing sales and warranty data from the Warranty Data Warehouse in Excel.

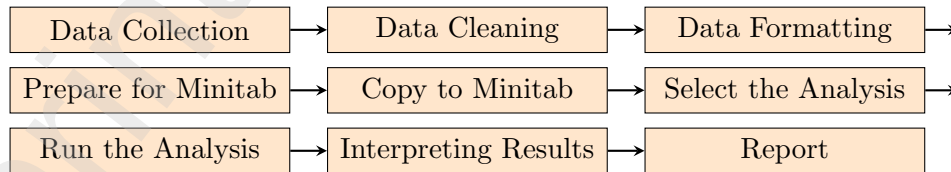


Fig. 1: Reliabilty Aanalysis in Minitab after Data Preprocessing in Excel

After collecting sales and repair data from the database, the next steps involve data cleaning, where duplicate or outlier data is removed. The de-

cision to conduct daily or monthly analyses is then made. In the data formatting stage for daily analysis, attributes such as usage days, breakdowns, and observations discontinuations are added to each vehicle's information. For monthly analysis, the data is aggregated into a Nevada Chart format by classifying it into calendar months. Several manual tasks during data preprocessing, such as attribute handling, warranty period entry, classification into breakdowns and observation discontinuations, calculation of usage days and mileage, and handling of single/multiple failure modes, can lead to numerous human errors. The prepared data is then copied to Minitab, and for daily analysis, the analysis is conducted with the choice of right-censored observations. For monthly analysis, arbitrary observation discontinuity analysis is chosen, and reliability analysis is performed. In the case of arbitrary observation discontinuity, the data format requires modification through the warranty data preprocessing menu. This involves specifying the start and end of the interval, as well as the format of breakdowns and observation discontinuities. Interpretation of graphs and results is done, and if additional analysis is needed, the process is repeated from data collection to interpreting results. This manual preprocessing and analysis process can introduce significant errors, and for quality engineers who need to make rapid judgments, it imposes many constraints, especially when the analysis target changes.

Successful development requires the integration and effective management of key elements such as clear goals, leadership, active participation from stakeholders, and efficient management of time and resources, including risk management. In this paper, we propose a series of processes for designing a digital dashboard that can support a range of quality improvement activities, starting from the basic information analysis required by a quality engineer for claims related to automotive components. The processes extend to in-depth reliability analysis and testing, aiming to provide essential information and insights to facilitate a comprehensive approach to quality enhancement in the field of automotive parts.

The main contributions of this paper are as follow:

1. The menu settings and layout, easily understandable terminology, and essential visualization charts for quality engineers were developed through numerous surveys, interviews, and multiple rounds of beta version testing, with input from a dashboard developer who also works as a quality engineer in the field.

2. Proposing an algorithm to handle warranty analysis target selection, data exploration, reliability analysis and test result analysis, warranty cost prediction, and validation of improvement measures all in one dashboard for quality engineers.
3. Expanding the reach of reliability engineering by handling cumulative failure probability distribution, development of accelerated test models, prediction of field claim rates through improvement test, and reliability validation tests for reliability target, all on one dashboard screen.

The organization of this paper is as follows: Literature Review section outlines warranty prediction, reliability analysis using reliability software, and quality dashboards, along with the limitations of these approaches. Additionally, the section mentions the focus that distinguishes this paper from these methods and what it aims to explain. Requirement section presents both the current issues with the claim information system and the key requirements of the dashboard proposed in this paper. The composition of a dashboard section elaborates on the form of the analyzed data, dashboard menu configuration, and considerations of UI usability. Each menu's feature section delves into various aspects such as warranty aggregation visualization, reliability analysis, warranty claim and cost prediction, improvement effect analysis, advanced analysis, and report download. Cost Reduction Effect section presents the opportunity cost gained from using the dashboard. The paper concludes with a summation of the findings and suggestions for future research.

2. Literature Review

- a) **Warranty Prediction and Statistical Modeling.** To achieve more accurate warranty predictions, numerous studies have been conducted. The studies that predicted warranty based on statistical modeling are as follows. [Kleyner and Elmore \(2022\)](#) provided a quantitative assessment of the warranty data maturity process and explained it through several case studies of return data in the field. They demonstrated how analysis results evolve as more service time elapses. [Babakmehr et al. \(2023\)](#) presented a data-driven framework for predicting automotive warranties, using Cox regression models to compensate for reporting delays. They explained the rush phenomenon at the warranty expiration point using adjusted warranty trends from historical data and considered a sales

delay model to predict the sales patterns of produced vehicles. Chehade et al. proposed a Conditional Gaussian Mixture Model (CGMM) using conditional Bayesian inference as a warranty prediction method (Chehade et al., 2022).

Recently, there has been a growing trend in research focusing on warranty prediction based on machine learning and deep learning. Ferencek et al. (2020) proposed a deep learning model for predicting warranty rates during the warranty period based on a dataset obtained from a home appliance manufacturer. Shokouhyar et al. (2021) suggested improving the accuracy of daily warranty claim predictions by incorporating additional social media data alongside company warranty data. Li et al. (2022) leveraged the advantages of models commonly used in machine translation and classical survival models to extract features from long time series and presented a model to predict the system's state over the entire lifecycle. Xu et al. (2023) proposed an analysis algorithm that combines probability distribution and deep learning.

The above-mentioned papers introduce specially designed statistical modeling to address the challenges of warranty claim predictions or propose methods using machine learning and deep learning techniques with manufacturing system data or social media data to enhance prediction accuracy. However, due to the necessity of a substantial training process with a large amount of historical or related data, for field quality engineers who require quick analysis results, the constraint of computing time and inconvenience in the data preparation process often outweigh the benefits of increased accuracy.

Therefore, this paper focuses on developing task customized algorithms that allow quality engineers to make quick judgments with minimal data, prioritizing practical utility in the field over the accuracy of warranty predictions.

- b) **Reliability Analysis Tools and Software.** Various methodologies for warranty or failure prediction fall within the category of reliability. Reliability not only aids in predicting potential warranty costs in the future but also assists in analyzing the types and causes of failures and understanding the forms of failures. Furthermore, reliability is closely associated with equipment availability and maintenance.

Many commercial statistical software products, such as Minitab, SAS, Weibull++, are utilized in reliability analysis (Kleyner and Elmore, 2022). Talkit et al. (2023), for instance, employed reliability analysis for scheduling maintenance using relevant software. Ray and Zhao (2022); Patil et al. (2022) estimated optimal distributions, Kleyner and Elmore (2023) utilized reliability demonstration tests, Patil (2022) applied reliability analysis for proactive failure management and improvement systems, and Liu et al. (2022) used Weibull distribution-based models for predicting automotive warranty data.

Furthermore, many people have utilized programming languages such as R or Python for reliability and warranty prediction. Teköz (2022), for instance, employed it for monthly warranty claim predictions in the service groups of the automotive sector. Pires et al. (2023) used R for classifying basic and validation models in the context of automotive warranty prediction. To model the technical availability of the system, Szkutnik-Rogoż et al. (2023) developed a mathematical model and assumptions for the actual process of transportation using a program.

Despite the advancements in reliability analysis software and the utility of packages provided by programming languages, familiarizing quality engineers with these tools and statistical theories remains a significant obstacle to expanding their usage effectively.

Therefore, this paper focuses on proposing a series of flows and configuration algorithms that enable quality engineers to analyze reliability and assess the effectiveness of accelerated test results for improvements easily on a single screen, without the need for separate software or direct use of programming languages, specifically for reliability analysis.

- c) **Data Quality and Automation.** When defining data quality as the extent to which data meets user requirements and expectations, a lack of accuracy in data analysis can lead to corrective activities, customer losses, and missed opportunities for organizations (Olson, 2003). In particular, human fatigue, inattention, or bias can be associated with serious data input errors, potentially resulting in completely incorrect judgments (Barchard and Pace, 2011). To reduce human errors, it is necessary to create and communicate standards defining the format, structure, and meaning of the data (Ladley, 2019). Additionally, separate data validation checks should be implemented to prevent or detect

errors such as missing values, outliers, duplicated values, and incorrect formats (Rahm et al., 2000).

It is well-known that transitioning manual tasks to automated processes in data analysis can reduce the risk of human errors and enhance operational efficiency and stability (O'Connor and Miller, 1987). Batini et al. (2009) introduced a methodology to enhance data quality, responsibility definition, error cause analysis, solution identification, process redesign, and improvement monitoring. de Paula Vidal et al. (2022) highlighted that by supporting decision-making in the company's inventory management through automated and systematic means via the management dashboard. Previous researchers commonly share the view that an automated dashboard tailored to the specific goals is needed to create a shared vision, values, and objectives related to quality.

In the aspect of the warranty and reliability dashboard, Gitzel et al. (2015) proposed a data quality dashboard for reliability data. Annadurai (2023) proposed an automated dashboard for efficiently analyzing a large number of products in a short period during warranty analysis.

However, the dashboards or automated analysis systems shown in the above paper are limited to a part of reliability analysis, and they do not capture the overall process related to quality engineer's claim improvement.

Therefore, this paper focused on proposing the design of a dashboard that allows quality engineers to easily execute a series of processes, including identifying key field issues, root cause analysis, establishment of improvement measures, and assessment of their effectiveness, all on a single screen. The emphasis is not limited to reliability analysis and change, but extends to showcasing and addressing major issues in the field.

3. Dashboard Requirements

Many companies currently possess claim information analysis systems to aggregate and visualize claim statuses, making field information easily accessible to members. However, if the dashboard is limited to exploratory data analysis (EDA), it may be insufficient from a risk management perspective. Simple visualization without sufficient statistical techniques has limitations

in perceiving patterns and trends hidden in the data, potentially leading to biased judgments and incorrect decisions. The limitations of such analyses can arise for various reasons. Firstly, if the goals regarding a company's warranty claims are not specific or management is inadequate, members may not see the need for more advanced analysis. Secondly, insufficient prior knowledge and effort from the client may result in inadequate requirements being communicated to the developer. The client must clearly articulate the purpose and objectives of the dashboard, requirements from user groups, data to be displayed, design and layout requirements, user interactions, and necessary functionalities before communicating them to the developer. Thirdly, the lack of competence and ongoing support from the developer can be an issue. If the developer lacks the ability to implement software for the requested features, the dashboard may be designed differently than initially intended. Moreover, if the development period is short or there is a lack of financial resources, the reliability may be compromised, and modifications and enhancements may be difficult, ultimately leading to abandonment.

The design guides provided by companies selling dashboard templates indirectly shed light on several elements that a warranty analysis dashboard should encompass. For instance, Qlik company advises the following: 1) Know your audience, 2) Choose the right type of dashboard, 3) Understand key metrics, 4) Tell a story with data, 5) Choose the right type of chart, 6) Apply essential dashboard UI design principles, 7) Iterate and evolve continuously. Similarly, Geckoboard, a company similar to Qlik, recommends the following: 1) Clearly define your goals and include only the essential elements, 2) Consider data-ink ratio, 3) Round numbers, 4) Use the most effective visualizations, 5) Group related metrics, 6) Maintain consistency, 7) Show hierarchy using size and position, 8) Provide context for numbers, 9) Use clear labels for users, 10) Remember it's for people, 11) Keep evolving the dashboard ([geckoboard, 2023](#)). Additionally, Bernardita Calzon, in a contribution to Datapine, explains what dashboards are and discusses the necessary features for strategic, tactical, operational, and analytical dashboards. She outlines the conditions a dashboard should meet for various analysis purposes and provides examples ([Bernardita Calzon, 2023](#)).

Combining these insights, a warranty analysis dashboard should meet the following criteria.

1. There must be a clear purpose. This dashboard manages the risk of

field claims by analyzing field warranty data in depth. Simultaneously, it identifies and improves components that need concentrated improvement during new vehicle development, aiming to achieve field warranty targets in advance during the development phase.

2. Defining users is indispensable. The primary users of this dashboard are quality assurance engineers responsible for field warranties and development quality engineers overseeing the quality of development vehicles. Additionally, engineers responsible for reliability testing and design are also included.
3. The menu should focus on providing an intuitive experience by addressing the information and functions needed by the main user groups of the dashboard. This dashboard is designed with a user-centered approach, offering insights into current and future claim predictions, warranty costs, identifying key failure patterns, and evaluating the effectiveness of improvement measures.
4. Reflecting user requirements is essential. The menus and dashboard output information, created based on user requests and preferences, enhance user satisfaction, improving the dashboard's usability. The tables and charts should not only provide new insights but also be intuitive and familiar, suitable for communication and reporting without modification. This dashboard underwent multiple test versions to ensure the fulfillment of user requirements.
5. A user-centric UI is essential. It assists users in easily understanding and navigating the dashboard. This dashboard is structured with panels following the analysis flow, minimal operation, freedom in choosing chart colors, dynamic chart configurations, conducting analyses of various components on multiple screens, and allowing report downloads, reflecting various user interface elements.
6. Easy program installation is emphasized. If the installation process is complex or cumbersome, it may take time for users to utilize the dashboard, potentially hindering operational efficiency. This dashboard allows individual installation or connection via a server. Using the dashboard through the server without individual installation is advantageous for security, maintenance, and updates.
7. Leadership and promotion to facilitate use are essential. When intro-

ducing the dashboard, it is necessary to report to the management about the purpose and usefulness of the dashboard and obtain sponsorship. Management directives and demands enhance the practical usability of the dashboard. This dashboard also expanded its usability through management's directives and became an essential operational process.

4. The composition of a dashboard

4.1. The form of the analyzed data

The target dataset for the dashboard presented in this paper consists of sales information and repair information. The sales information includes unique identifiers such as the VIN (Vehicle Identification Number) and details necessary for in-depth analysis, such as the model, sales region, production and sale dates, and key powertrain specifications, as shown in [Table 1](#). The VIN is a 17-character alphanumeric code that follows ISO standards, identifying the manufacturing country and company, along with groups for vehicle characteristics and production serial numbers. The Model Year (MY) represents a model change in automobiles, indicating a change in model year without significant external alterations. The Sail Region (SR) signifies the sales region, usually divided by continent, where similar regulations apply. P.Date stands for Production Date, and S.Date represents Sail Date. EG denotes Engine, and TA denotes Transaxle, with the model name specified.

The repair information includes various attributes for vehicles that have undergone repairs, as shown in [Table 2](#). PN stands for Part Number, representing the specification information for the replaced component. It consists of a representative 5-digit number for the part group and the remaining 5 digits mixed with alphanumeric characters, indicating the applicable vehicle model and grade. SYS refers to the System to which the repair part belongs. SUP denotes the Supplier and is used for warranty analysis by supplier. Failure indicates the symptoms of the malfunction, providing more detailed specifics if necessary. Comment is used for entering detailed information such as customer complaints that are not adequately covered by standardized classifications, diagnostic content evaluated by the serviceman, and resolution status after parts replacement. R.Date represents the repair date, while MIL indicates the mileage traveled up to the repair point in kilometers. Cost includes repair costs, represented as the sum of part cost and labor cost. The

sales and repair datasets are linked by VIN, and ultimately, the attributes from the repair dataset are added to the sales dataset as columns corresponding to the VIN. The usage days are then calculated.

For each VIN, a binary attribute is added to indicate whether a failure occurred or not. '1' represents a failure occurrence indicator, while '0' indicates no failure. The usage days until the observation discontinuation are displayed as the warranty period days if the warranty period has passed; otherwise, it shows the usage days until the observation date

VIN	MY	SR	P_Date	S_Date	EG/MOTOR	TA/REDUCER
XXXXXXXXXXXX94347	XXMY	USA	20XX0420	20XX0630	A E/G	A TA
XXXXXXXXXXXX94784	XXMY	KOREA	20XX0412	20XX0424	B E/G	B TA

Table 1: Sale Data Set

VIN	PN	SYS	SUP	Failure	Comment	R_Date	MIL	EA	Cost
XXXXXXXXXXXX94347	XXXXXXXXXXXX	XXX	A	NOISE	XXX	20XX0101	XXX	1	XXX
XXXXXXXXXXXX94784	XXXXXXXXXXXX	XXX	B	LEAK	XXX	20XX0102	XXX	1	XXX

Table 2: Repair Data Set

4.2. Dashboard Menu Configuration

The primary users utilizing this dashboard are quality engineers. The key preferred menu items, as received through a survey conducted by the pre-quality engineer, are shown in the following order in Fig. 2.

Through open-ended responses, the gathered requirements emphasized the desire for accurate and prompt prediction of future claims trends, as well as a more accessible analysis of the effectiveness of improvement measures, with a preference for simplified statistical terminology. Additionally, there was feedback indicating the need for enhanced usability in obtaining results for reliability analysis.

While surveys provide an overview of general requirements, direct consultation with quality engineers is essential for detailed menu design. Below is a representative persona developed based on the interview results. Personas are virtual user profiles that represent specific user groups, aiding designers and developers in designing and developing user experiences (Banerjee, 2004). By translating the identified requirements into menu items and determining their order, this persona assists in making informed decisions for dashboard design. We provide a snippet of the persona created through the interviews.

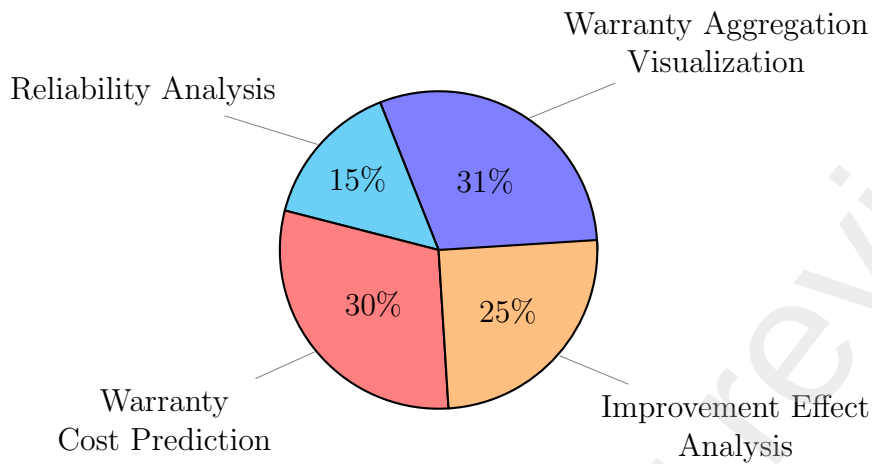


Fig. 2: Quality Engineers' Top Menu Preferences

Engineer A is responsible for the development quality of subsystem X. When developing a new vehicle, if the claimed costs of the responsible subsystem rank above a certain threshold compared to the existing vehicle, Engineer A is accountable for reducing the failure rate in the developed vehicle below a certain level. Before releasing the drawings, it is necessary to explain the claims on the existing vehicle to the designers and collaborators, and discuss the key improvement areas. Since the systems in the developed vehicle are often similar to those in the existing vehicle and claims tend to carry over, an objective analysis method is required, as simple charts of production and repair rates on a monthly calendar basis may change significantly after a few months.

While statistical analyses such as reliability analysis are known to provide objectivity, they require separate training and the use of complex programs, causing a significant burden. It would be beneficial to have a system that is easily understood in everyday terms and provides conclusions with just a few mouse clicks. Additionally, the ability to distinguish whether a problem is due to design issues or process defects when identifying the root cause would be valuable, as it can prevent conflicts and resource wastage during departmental decision-making.

It is also necessary to assess the effectiveness of improvement measures. Often, failure modes with many claims have multiple improve-

ment measures in place, requiring quantitative analysis of the improvement effects. Reports are needed for discussions with other departments or suppliers, and the time spent on document creation is not negligible. Efficiency in work would be enhanced if the required plots and tables could be generated directly as documents.

The flowchart depicted in Fig. 3 has been created based on a survey of various engineer preferences, outlining the dashboard design.

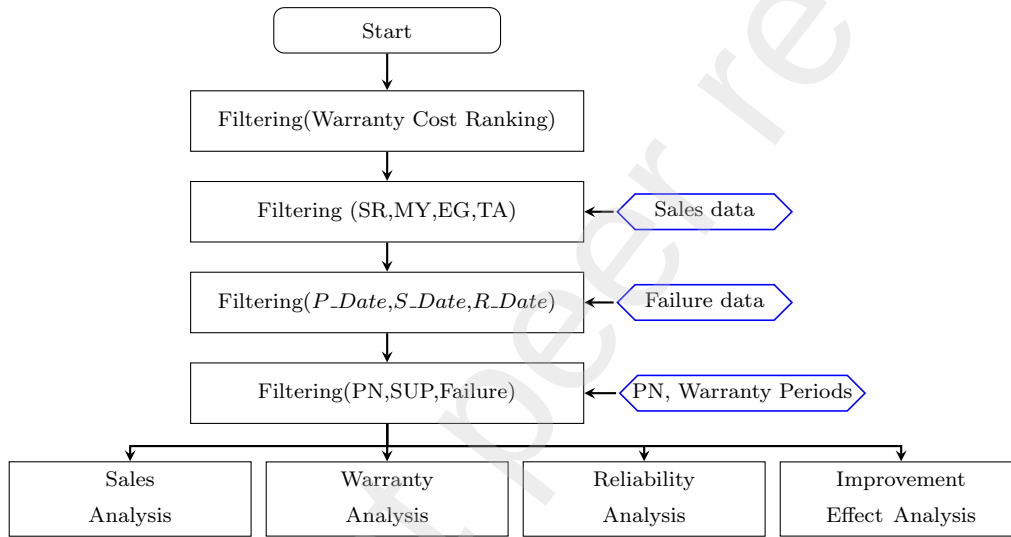


Fig. 3: Flowchart of panels in the dashboard

4.3. Considerations for UI usability

The flow of panel processing for analysis tasks is designed to proceed from left to right, following the natural workflow. Menus that users can directly access, such as improvement effect analysis, are designed to be processed separately, regardless of their order. Furthermore, menus primarily used and needed by quality engineers are defaulted for quick and easy access. In-depth or complex functionalities are hidden for users who do not require them. Specialized terms used in statistics or reliability engineering are avoided as much as possible, and replaced with simpler terms. For example, instead of "testing the equality of shape parameters in the cumulative failure probability distribution," the expression "Are the slopes of the fitted lines similar?" is used.

To minimize the inconvenience of searching for additional information or statistical explanations, items requiring additional clarification are provided through mouse popover functionality. Typing is minimized to encourage users to execute commands primarily through mouse clicks. For charts and tables, it is recommended to maintain a similar framework and color scheme for visual consistency. If titles, legends, or axis labels are unnecessary, hiding or deleting them is advisable.

One essential element in dashboard design is user testing. Once a prototype is created, it is vital to gather feedback from target users after they have used the dashboard. This feedback is pivotal for the iterative process of improving the existing dashboard.

5. Each menu's features

The dashboard is created using R software, and specifically, calculations and plots related to reliability utilize R libraries. This library can handle common life distributions used in reliability analysis

5.1. Warranty Aggregation Visualization

Basic Chart. The analysis of the claimed cost and failure rate of the worst components during the target period of the analyzed vehicles determines the level and the necessity for detailed analysis of the components under consideration. Upon selecting the identified components for analysis, charts are provided, offering insights into the monthly sales and production quantities, production-monthly repair counts and failure rates, sales-monthly repair counts and failure rates, as well as the warranty status based on usage days and mileage. These charts serve as a fundamental understanding of the warranty situation for the selected component.

Users can prioritize improvement efforts by comprehensively understanding the overall pattern of failures through charts such as Fig. 4 and Fig. 6. These present a bar graph illustrating the monthly trends in repair counts for both production and sales, providing an overview of the monthly trend for overall claims. Also these include pie charts depicting the regional and failure-specific market share, as well as scatter plots and histograms illustrating the relationship between usage days, mileage, and failure occurrences. Providing users with a visual representation of the distribution of failure incidents helps them prioritize issues that require improvement. Pareto principle, applied in

defining phenomena and determining key control factors, is relevant in this context (Rai and Singh, 2009).

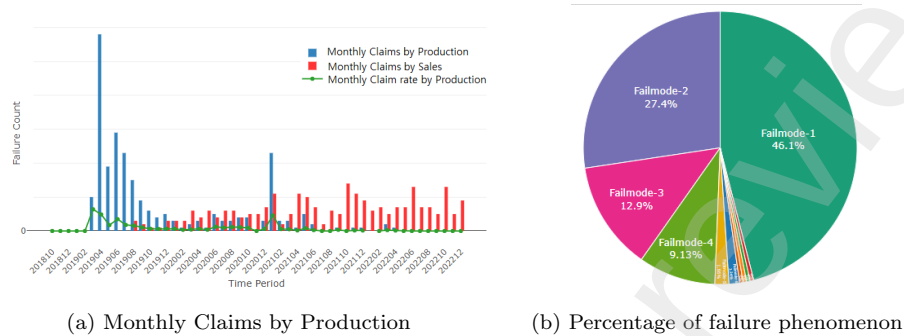


Fig. 4: Visualization for sales and failure data

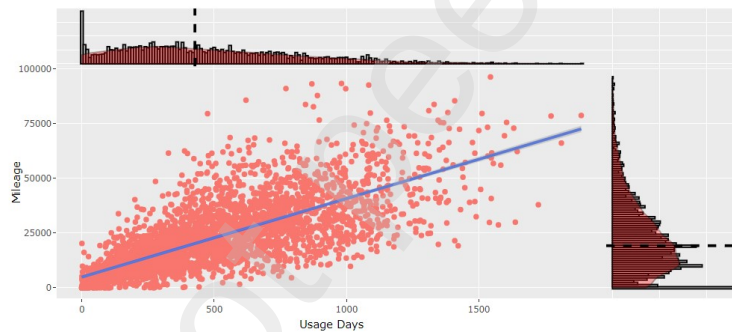


Fig. 5: Scatter plot of days used and mileages

Text filtering and Word Cloud. When the classification of failure phenomena lacks granularity or is ambiguous, or when there are doubts about the classification choices made by service engineers, the dashboard is configured to address such situations in three ways:

- Extraction of VINs Based on Detailed Technical Descriptions

VINs are extracted based on the detailed technical content that matches the analysis conditions, allowing for aggregation and reliability analysis.

- Scatter Plot of Failure Phenomena Based on Mileage/Usage Days

Similar failure phenomena are combined by examining the scatter plot of classified failure phenomena. For the scatter plot in Fig. 6a, a bi-

variate normal distribution is assumed, and contours for the 95% confidence interval are provided. When contours are similar, the related failure phenomena are treated as similar.

- Word Cloud

Additionally, selecting VINs solely based on detailed technical descriptions may be impractical when dealing with a large number of failures. To enhance this process more effectively, the dashboard provides a 'Word Cloud' feature, as illustrated in Fig. 6b. This feature filters based on frequently extracted VINs words, allowing for a more feasible approach to selecting relevant VINs.



Fig. 6: Similar Fail and Word Cloud

Incidents Per Thousand Vehicles. In the case of automotive OEMs, The number of Repairs Per Unit (RPU) and Cost Per Unit (CPU) indices are used as key Quality Key Performance Indicators (KPIs) (Babakmehr et al., 2023). RPU is sometimes referred to as IPTV (Incidents Per Thousand Vehicles) or R/1000, representing the number of failure per 1000 vehicles.

Fig. 7 illustrates the trend of Cumulative Failure Rates per 1000 units based on months of usage for each production month. Typically, vehicles in the early stages of sales exhibit higher failure rates, but over time, continuous improvements result in a decreasing slope. If a consistently high level is observed, it is necessary to consider potential design issues. In case of abnormal increases midway, an investigation into changes in 4M (Man, Machine, Material, Method) factors should be conducted. For easier discernment, utilizing IPTV charts on a production or sales monthly basis, as shown in Fig. 8, can aid in warranty management and root cause identification. In Fig. 7 and Fig. 8, the production for January 2021 shows a significant increase compared

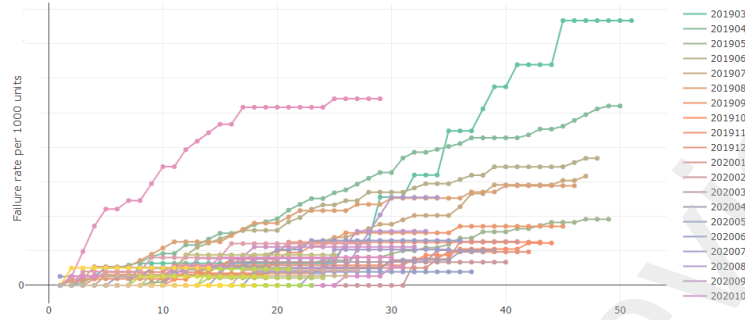


Fig. 7: Total IPTV by production month

to other months, suggesting a need to investigate what changes occurred in that specific period

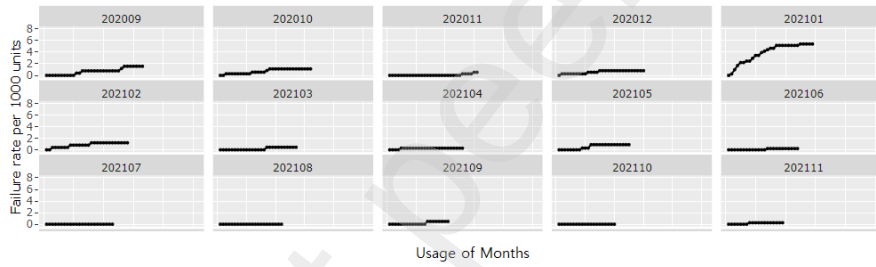


Fig. 8: Individual IPTV by production month

5.2. Reliability Analysis

The independent variables are the failure time and mileage, and the estimated distributions include the Weibull and log-normal distributions commonly used in warranty analysis. The default selection specifies the failure time as the independent variable and sets the Weibull distribution as the estimation distribution, with x representing the failure time in days of usage. The Weibull distribution has the following probability density function, as expressed in (1).

$$f(x; \beta, \eta) = \frac{\beta}{\eta} \left(\frac{x}{\eta} \right)^{\beta-1} \exp \left[- \left(\frac{x}{\eta} \right)^{\beta} \right], \quad x > 0 \quad (1)$$

where: β is the shape parameter and η is the scale parameter.

In cases where there is a significant amount of censored data, such as warranty data, the Maximum Likelihood Estimate (MLE) method is preferred for parameter estimation (Genschel and Meeker, 2010). Defining the data as $x_1, \dots, x_n$ and assuming that x_i is an independent variable for all i , (1) can be expressed as (2).

$$f(x_i; \beta, \eta) = \frac{\beta}{\eta} \left(\frac{x_i}{\eta} \right)^{\beta-1} \exp \left[- \left(\frac{x_i}{\eta} \right)^{\beta} \right], \quad x > 0 \quad (2)$$

Warranty claim data commonly contains a significant amount of censored data. When defining the probability variable for failure data as $D_f = \{t_1, \dots, t_n\}$ and the sample for censored data as $D_c = \{c_1, \dots, c_r\}$, the probability of the survival function can be expressed with the following likelihood function.

$$L(\theta; D_f, D_c) = \prod_{i \in D_f} f(t_i; \theta) \prod_{j \in D_c} R(c_j; \theta) \quad (3)$$

Where, $R(c; \theta)$ represents the reliability function, which is equal to $1 - F(t; \theta)$. For the Weibull distribution, the likelihood function of the survival function is given by (4).

$$L(\beta, \eta; D_f, D_c) = \prod_{i \in D_f} \frac{\beta}{\eta} \left(\frac{t_i}{\eta} \right)^{\beta-1} \exp \left[- \left(\frac{t_i}{\eta} \right)^{\beta} \right] \prod_{j \in D_c} \exp \left[- \left(\frac{c_j}{\eta} \right)^{\beta} \right] \quad (4)$$

log-likelihood function is:

$$\log(L(\beta, \eta; D_f, D_c)) = \sum_{i=1}^n [\log(\beta) - \beta \log(\eta) + (\beta-1) \log(t_i) - \left(\frac{t_i}{\eta} \right)^{\beta}] - \sum_{j=1}^r \left(\frac{c_j}{\eta} \right)^{\beta} \quad (5)$$

The unknown parameters α and β are obtained using numerical optimization methods since a closed form solution for the maximum likelihood estimate (MLE) of the log-likelihood function cannot be derived. In this dashboard, the Nelder and Mead method (Nelder and Mead, 1965), which

operates without derivatives, is used as the default optimization approach. Additionally, users have the option to choose quasi-Newton methods such as 'BFGS' or the conjugate gradients method ('CG').

For a quality engineer, if there is an excessive occurrence of claims for a particular component, improvement measures need to be formulated. Especially for fundamental improvements, field reliability analysis and claim predictions through accelerated life testing of improvement specifications are often essential. While the dashboard aims to handle this process on a single screen, if the user's knowledge of reliability engineering is limited, or if there are complex procedures, the utility of the dashboard may be limited.

The results should be presented in simple terms for easy understanding of reliability, and the desired output should be achievable with minimal processing. Fig. 9 illustrates the implemented reliability improvement process in the dashboard. Here, ALT refers to Accelerated Life Test, and RDT stands for Reliability Demonstration Test. Fig. 10 displays the CDF (Cumulative Distribution Function) based on the assumed Weibull distribution for each stage of the process. It includes Warranty, similarity testing with accelerated test results, extraction of acceleration factors through comparison of pre-improvement and post-improvement tests, and prediction of field claims after improvement compared with pre-improvement claims.

Additionally, for the Reliability Demonstration Test (RDT), the dashboard provides an automatic selection of test time and sample size for the zero-failure test mode using the Weibayes zero failure test plan, designed according to (7). Assuming independent and identical tests under the same conditions, the relationship between the maximum number of failures r in n tests and the lower confidence level CL is described by (6). The probability of no failures in n trials is expressed by (7). If the warranty target failure rate is provided, the acceleration factor obtained from the accelerated test model is used to convert it into the test target life. Then, with the minimum test time for demonstrating reliability and the minimum testable time, the required sample size is calculated if allocated sample size is provided.

$$P(x \leq r) = \sum_{x=0}^r \binom{n}{x} (1-R)^x R^{n-x} \leq 1 - CL \quad (6)$$

$$1 - CL = R^n \quad (7)$$

Additionally, for advanced analysts, non-parametric analysis options such as Kaplan-Meier analysis, Hazard analysis, 3-parameter Weibull analysis, and competing analysis using 3-parameter Weibull are provided as options.

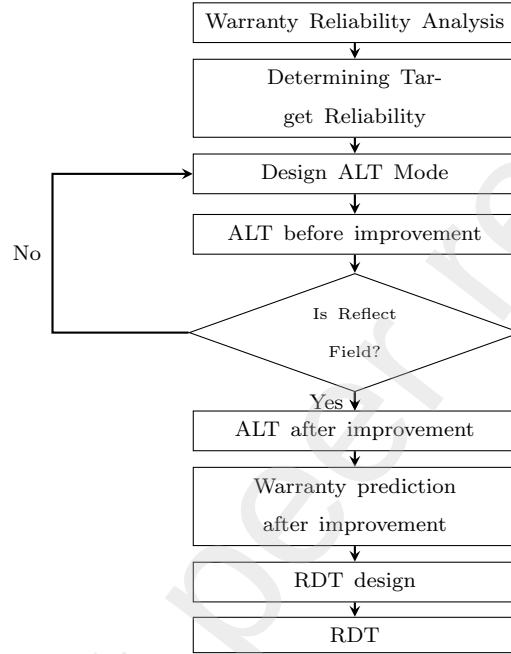
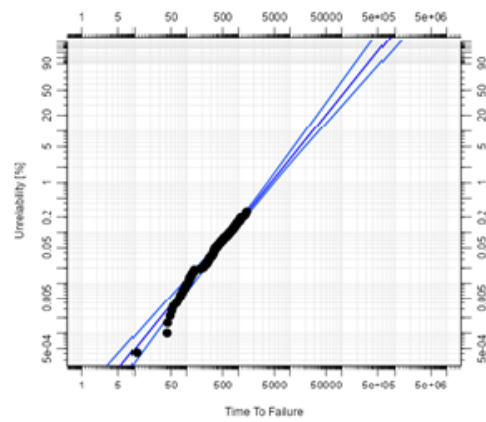


Fig. 9: Reliability Analysis and Improvement process

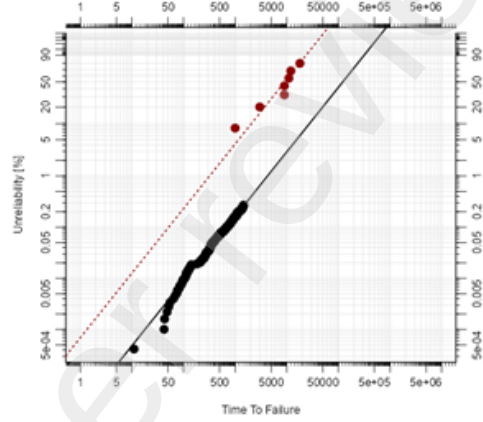
Reliability Analysis of Mileage. In cases where the dataset is based on days, as shown in Table 1 and Table 2, it is treated as right-censored. If the data is based on a monthly basis, it is treated as interval-censored. While the usage days of vehicles with observed failures can be calculated, the mileage may vary significantly based on factors such as vehicle class, operating region, usage months, and users, even for the same duration. Fig. 11 presents a histogram of daily mileage for the same recalled vehicles in the United States and Korea, illustrating regional differences.

Estimating mileage involves various methods, with three common approaches:

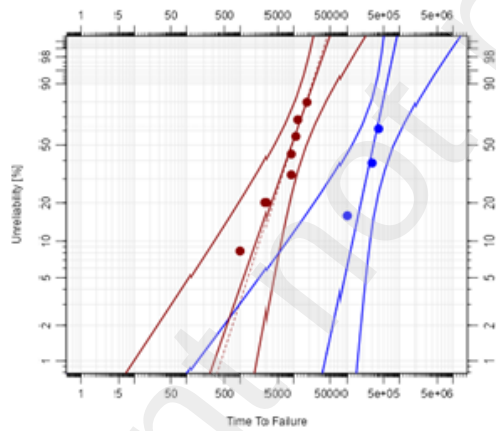
- 1) Sampling Vehicle Measurement: Measuring the mileage of a sample of vehicles.
- 2) Analysis of Mileage Information for Warranty Claims of Vehicles: Analyzing mileage information from vehicles that experienced failures.
- 3) Utilizing Onboard Communication Systems: Using the communication systems



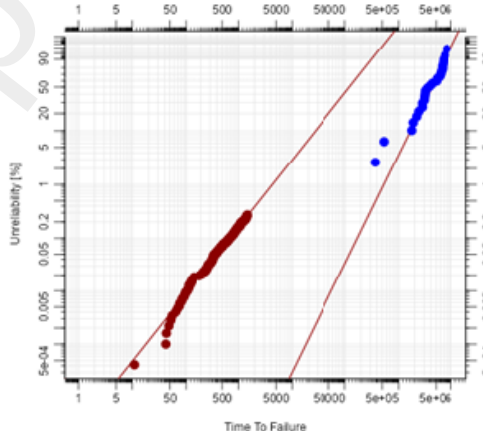
(a) CDF of Warranty data



(b) CDF of the before improvement testing and Warranty data adjusted for the same shape parameter



(c) CDF of tests of the before and after improvement



(d) CDF of Warranty data before and after improvement.

Fig. 10: Reliability improvement and Warranty prediction flow ($a \rightarrow b \rightarrow c \rightarrow d$)

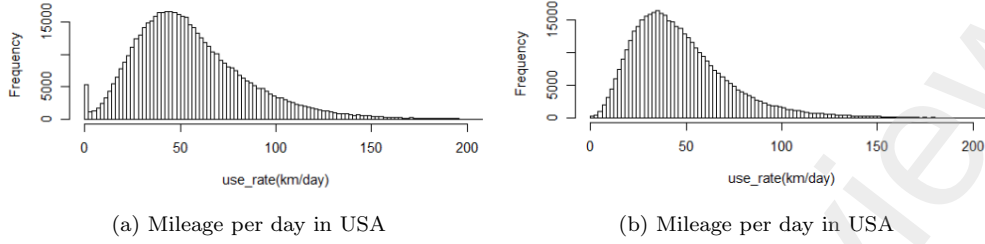


Fig. 11: Histogram of mileage per day

onboard vehicles to collect mileage data. An exemplary case of analyzing mileage from sampled vehicles is presented by [Lu \(1998\)](#), who assumed a log-normal distribution based on mileage data measured from North American vehicles and analyzed warranty information.

To analyze mileage information from faulty vehicles, a significant amount of failure data is required. To prevent overfitting, it is recommended to estimate mileage from a large dataset covering failures across various vehicle types, providing tens of thousands of pieces of information.

While utilizing communication systems for mileage data collection offers the advantage of obtaining abundant information, it has limitations as it is restricted to vehicles connected to the communication system.

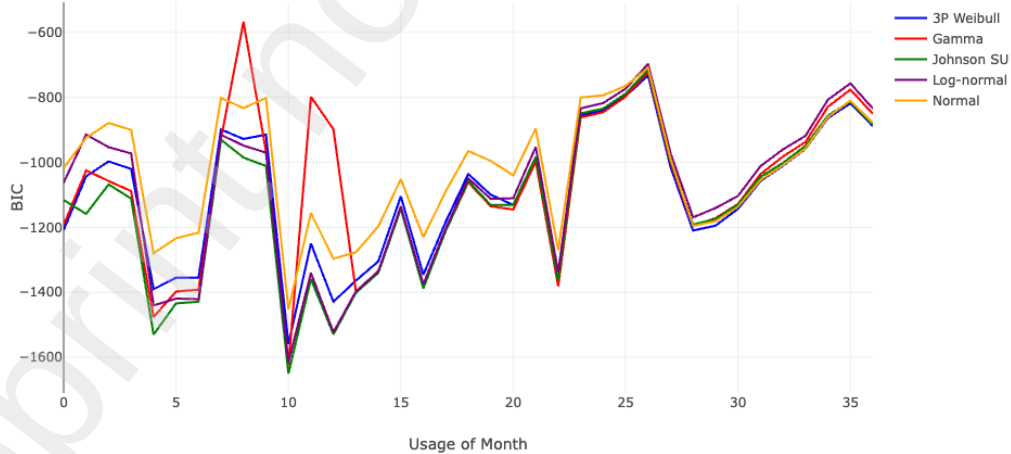


Fig. 12: BIC for the distribution of the average daily mileage over the months of use

In the dashboard, the mileage estimation utilized the overall mileage information for Warranty Claims of Vehicles. Fig. 12 illustrates the graph of the average daily mileage over the months of use for approximately 200,000 SUV vehicles sold in Korea. The graph also shows the Bayesian Information Criterion (BIC) for various estimated distributions. Overall, the Johnson SU distribution exhibits consistently lower BIC values across several months of use, suggesting it as the optimal fit. However, apart from the initial months, there is little significant difference observed between the log-normal distribution, Weibull 3-parameter distribution, and Johnson SU distribution. (8) represents the probability density function (PDF) of the Johnson's SU distribution.

$$g(x; \delta, \lambda, \gamma, \xi) = \frac{\delta}{\lambda\sqrt{2\pi}} \frac{1}{\sqrt{1 + (\frac{x-\xi}{\lambda})^2}} e^{-\frac{1}{2}(\gamma + \delta \sinh^{-1}(\frac{x-\xi}{\lambda}))^2} \quad (8)$$

In computer simulations, finding solutions for four parameters - Scale Parameter (δ), Shape Parameter (λ), Shape Parameter (ξ), and Location Parameter (γ) - is not an easy task. Additionally, the substantial dispersion of BIC values across months of use raises doubts about assuming a particular distribution. For these reasons, in this dashboard, the mileage of the vehicles with observed discontinuity was assumed using a linear regression equation for the median mileage across months of use.

5.3. Warranty Claims and Costs Prediction

Prediction warranty claims and costs enables the preparation for potential future financial burdens, allowing for advance budget planning. It assists in decision-making processes such as recalls, campaigns, and determining the necessity for warranty extensions. Effective management of service inventory is facilitated, and improvements in product design and defect resolution can enhance customer loyalty in the long run.

The transformation of the warranty dataset. In this dashboard, claims are assumed to be aggregated on a monthly basis. Such data can be represented as shown in Table 3. For a given calendar month τ , our goal is to predict the total number and cost of warranty claims that may be reported over the next k months. In Table 3, the number of units delivered to end-users in month i and the number of claims reported in month j are denoted as $d_{i,j}$, while

MonS	Shipments	Calendar months when claims are received							
		1	2	...	τ	$\tau + 1$	$\tau + 2$	...	$\tau + k$
1	n_1	$d_{1,1}$	$d_{1,2}$	...	$d_{1,\tau}$	$\hat{d}_{1,\tau+1}$	$\hat{d}_{1,\tau+2}$	...	$\hat{d}_{1,\tau+k}$
2	n_2		$d_{2,2}$	...	$d_{2,\tau}$	$\hat{d}_{2,\tau+1}$	$\hat{d}_{2,\tau+2}$	...	$\hat{d}_{2,\tau+k}$
...	...			...	...	...	...	...	...
τ	n_τ				$d_{\tau,\tau}$	$\hat{d}_{\tau,\tau+1}$	$\hat{d}_{\tau,\tau+2}$	...	$\hat{d}_{\tau,\tau+k}$
$\tau+1$	m_1					$\hat{d}_{\tau+1,\tau+1}$	$\hat{d}_{\tau+1,\tau+2}$	...	$\hat{d}_{\tau+1,\tau+k}$
$\tau+2$	m_2						$\hat{d}_{\tau+2,\tau+2}$	...	$\hat{d}_{\tau+2,\tau+k}$
...	...							...	...
$\tau + k$	m_k								$\hat{d}_{\tau+k,\tau+k}$

Table 3: Monthly Sales and Repair Aggregation

the predicted warranty claims are represented as $\hat{d}_{i,j}$. 'MonS' indicates the month in which the product is delivered to the end-user.

Through this, if we denote the number of right-censored observations at the boundary point t_i for $i = 1, \dots, k$ represented by the lifetime intervals as s_i , it is expressed as the difference between the sum of elements in the $(k + 1 - i)$ th row and the sales quantity in that row, as shown in (9).

$$s_i = n_{\tau+1-i} - \sum_{j=\tau+1-i}^{\tau} d_{\tau+1-i,j} \quad (9)$$

If the future periods $1, 2, \dots, k$ have shipments of $m_1, m_2, \dots, m_k$ respectively, and the warranty period is specified as W , the Expected Number of Failures (PNF) for the future period Δ is given by (10). Here, $R(t)$ is the reliability calculated using the Shape and Scale parameters obtained from (5), and I_C is an indicator function, which becomes '1' if condition C is satisfied and '0' otherwise.

$$PNF(\Delta) = \sum_{j=1}^k s_j \left(1 - \frac{R(t_j + \min\{\Delta, W - t_j\})}{R(t_j)} \right) I_{t_j < W} + \sum_{j=0}^k m_j [1 - R(\min\{\Delta + (1 - j), W\})] \quad (10)$$

Research on this topic can be found in (Wu and Akbarov, 2012; Stephens and Crowder, 2004; Kalbfleisch et al., 1991; Minitab, 2023). When presenting the results graphically, it is possible to represent the failure count estimated as a Weibull distribution from the existing warranty claims. However, it is more helpful to show the actual occurrences in the field. Combining the estimated future failure counts and costs from (10), the representation is shown in Fig. 13.

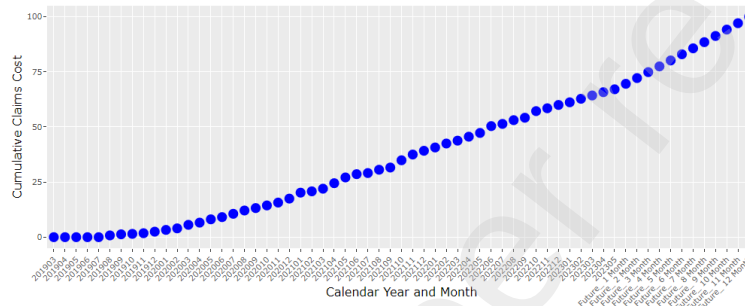


Fig. 13: Warranty Cost Prediction

5.4. Improvement Effect Analysis

When improvement effects are considered but prove to be insufficient over time, it often leads to cost wastage, decreased customer satisfaction, loss of trust within the organization, and potential legal issues such as recalls or product liability lawsuits. Analyzing the performance before and after the implementation of improvement measures allows for a quick and accurate comparison, enabling the identification of root causes of issues and deriving additional measures to prevent recurrence of similar problems. For user convenience, the system is configured with the option to select the improvement date in FRACAS (Failure reporting, analysis, and corrective action system), a system that combines direct input and quality improvement information. The result graphs are shown in Fig. 14. Fig. 14a compares the number of failures and claim rates before and after the implementation of improvement measures. Fig. 14b presents a comparison of the average IPTV values for production months before and after improvement. Cumulative failure probability estimation is provided in Fig. 14c, allowing observation of the trend in failure occurrence over time. Analyzing whether the improvement effects are

effective throughout the entire period or if there is no effect after a specific period helps assess the need for additional improvement measures.

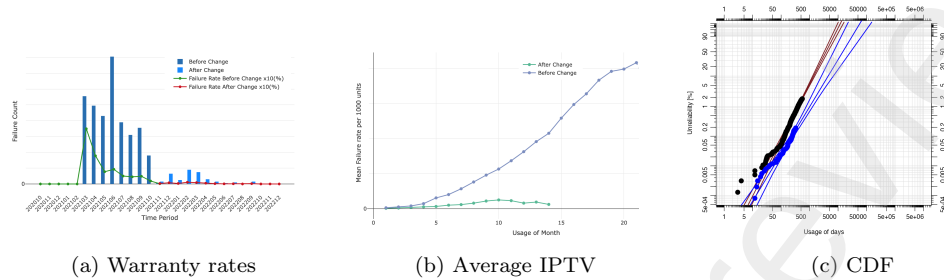


Fig. 14: Comparison of before and after improvement

5.5. Advanced Analysis

For quality or reliability engineers, advanced reliability analysis methods are provided. When multiple failure modes are present in reliability analysis, the system offers competing models by separating each failure. Fig. 15 illustrates different failure distributions based on a 200-day reference point. By separating the trend of each failure, it aids in individual predictions, as shown on the right side.

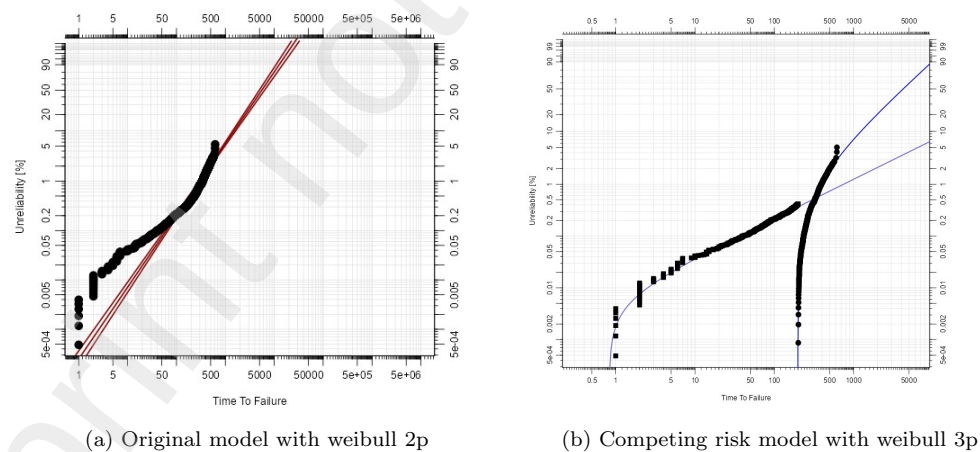


Fig. 15: Competing risk model

It also provides a menu for analyzing the results of accelerated life testing based on the failure times (cycles) and censored time (cycles) under

operational conditions. Users can input the target life and test stress under operating conditions. By using representative models such as the Arrhenius model and power model, it allows comparison of the target life under normal operating conditions and the extrapolation of the accelerated life testing results for the test subject, as shown in Fig. 16. It also calculates the Activation Energy and Power factor.

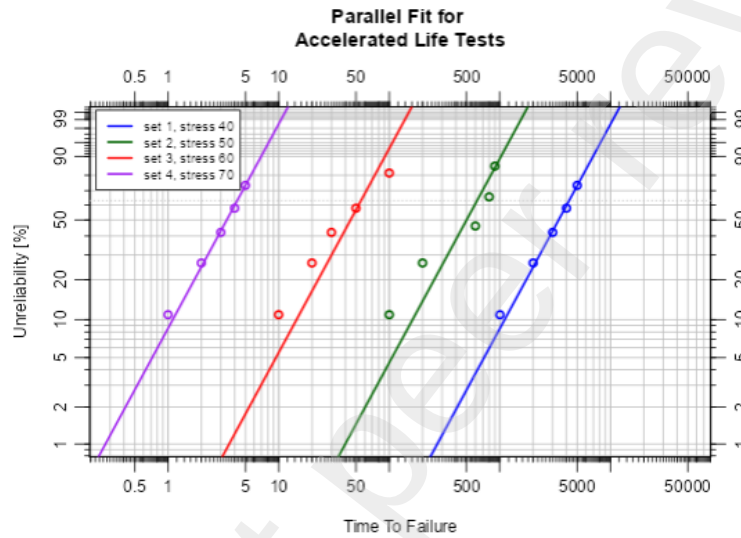


Fig. 16: Accelerated life test model

5.6. Report download

The reporting feature in the dashboard offers several advantages. By directly exporting the charts and tables from the dashboard into document format, it enables the quick and easy generation of data for decision-making. Setting up the format and titles for each sheet in advance, along with the ability to choose the output range, is programmed to significantly reduce the time required for report creation. The generated reports can be utilized for communication with individuals, teams, external stakeholders, or customers, enhancing collaboration and communication. Moreover, it aids in decision-making and strategic planning for the management, serving as a pivotal tool for data-driven decision-making and overall efficiency improvement within the company.

6. Cost Reduction effect

The cost improvement effect in the dashboard can be calculated using (11).

$$Cost = N \times \frac{AS \times n \times h}{w \times d \times k} \quad (11)$$

where:

N = The number of engineers performing quality analysis.

AS = The average salary of engineers conducting quality analysis.

n = The number of quality analysis tasks performed by one quality analysis engineer in a year.

h = The time required for one quality analysis task.

w = The number of working weeks in a year.

d = The number of working days in a week.

k = The number of working hours in a day.

Assuming that the average annual salary for a quality engineer is 60 thousand USD, and approximately 200 engineers perform analyses similar to the dashboard 20 times per year per person. Engineers work for 50 weeks a year, 5 days a week, 8 hours a day. Manual analysis, including analysis and report generation per session, takes an average of 16 hours, while using the dashboard requires less than 10 minutes.

According to the formula above, the cost for manual analysis is approximately 1.92 million USD per year, and using the dashboard incurs a cost of 0.02 million USD per year. The savings in opportunity cost, as profit, amount to 1.9 million USD annually.

7. Conclusion and future works

High quality is essential for corporate profitability and customer satisfaction. Accurate claims analysis is foundational for identifying and improving product defects, enhancing customer satisfaction, and preventing unnecessary costs.

This paper focuses on designing a digital dashboard for automotive components, emphasizing support for quality improvement activities. Dashboards play a key role in predicting claim trends through data analysis, enabling swift and accurate countermeasure formulation. Automated dashboards reduce human errors, enhancing operational efficiency and facilitating effective communication among stakeholders.

Developed through insights from literature, guidance from the manufacturing company, and multiple test versions involving interviews with field engineers, this dashboard addresses their demands in analysis and improvement activities. It considers UI aspects to design menus integral for reliability testing analysis, warranty cost prediction, and improvement effects analysis.

The utilization of this dashboard led to a reassessment of the effectiveness of various quality improvement activities conducted in the past. Additionally, it focuses on identifying components and failure patterns that require concentrated improvement during new vehicle development, monitoring the improvement progress. The enhanced analysis speed not only reduces opportunity costs but also leads to significant cost savings.

As part of future work, the first initiative aims to enhance accountability at procurement units by implementing a system that allocates warranty targets on the warranty dashboard. By introducing this system, we seek to establish a transparent framework that fosters effective management of warranties and ensures a focused and responsible approach within each procurement unit. For the second endeavor by leveraging deep learning on diverse datasets encompassing production, management, and vehicle sensor data, the intention is to develop a robust model that anticipates and prevents potential warranty claims associated with customer dissatisfaction.

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